

Range Ambiguous Clutter Suppression Based on Random Pulse Repetition Interval for Spaceborne Radar Systems

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Abstract—Space-time adaptive processing (STAP) technology can effectively suppress the main-lobe clutter of spaceborne radar and coherently integrate weak target signals over a wide area, thereby significantly improving the target detection performance of spaceborne radar systems. However, the presence of range ambiguity enhances the clutter degree of freedom and clutter power level, which degrades the clutter suppression capability of STAP and consequently compromises the overall target detection performance. To address this challenge, this paper proposes a novel approach that integrates random pulse repetition interval (RPRI) technology with non-uniform Fourier transform and reduced-dimension STAP techniques. By leveraging the randomization of pulse intervals, the ambiguous-range clutter corresponding to different pulses in a multi-pulse sequence, exhibits different characteristics, which prevents the coherent accumulation of ambiguous clutter energy across pulses. This mitigates the clutter spectrum spreading problem induced by range ambiguity. Nevertheless, this method inevitably elevates the base noise level, resulting in a reduction in signal-to-noise ratio (SNR). Therefore, it is particularly suitable for scenarios with high yaw angles and sufficiently high target SNR. Finally, simulation results are presented to validate the efficacy and application scope of the proposed method.

Keywords—Range ambiguity, clutter suppression, random PRI, spaceborne radar.

I. INTRODUCTION

Compared with ground-based radars and airborne radars, spaceborne radars have significant advantages such as a wide detection range, long-duration monitoring capability, resistance to occlusion, and unrestricted operation across national boundaries. These strengths have led to their extensive applications in environmental monitoring, territorial security, geological exploration, and other fields. However, the high-speed movement of satellite platforms and ground-surface reflections introduce strong spread clutter in spaceborne radar echoes. This clutter often obscures weak

and small targets, significantly degrading overall target detection performance. To address this issue, Brennan [1] proposed the Space-time adaptive processing (STAP) technique, which achieves clutter suppression and effectively enhances target detection capability. However, the independent and identically distributed samples required for this technique should be greater than twice the product of the number of transmitted pulses and the number of channels, which is difficult to be achieved in practice. Consequently, many near-optimal STAP techniques have been proposed, such as the extended decomposition filtering and adaptive processing (EFA) STAP algorithm proposed by R. Dippetto et al. [2]. This method uses adjacent Doppler channels of the target as auxiliary channels for clutter suppression processing, reducing the system dimension to three times the number of channels.

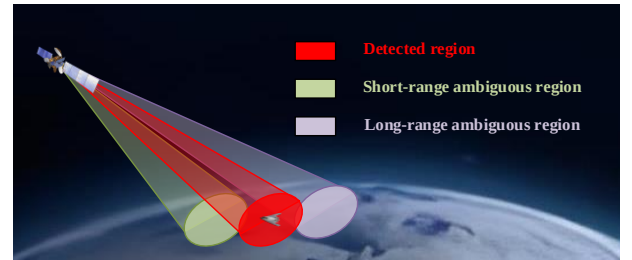


Fig. 1. Ketch map of range ambiguity.

Additionally, due to the operation of spaceborne radar platforms at higher altitudes and with specific orbital inclinations, the difference in surface line velocity caused by the Earth rotation and the satellite motion direction leads to the occurrence of the crab effect. This effect causes additional Doppler frequency shifts in the clutter signals. Furthermore, when spaceborne radar detects high-speed aerial targets at long distances, a higher pulse repetition frequency (PRF) is typically required to avoid Doppler ambiguity. However, this can result in range ambiguity, causing the received echo signals to be a superposition of reflections from multiple ambiguous regions, as shown in Fig. 1. It is important to note that the clutter characteristics in terms of space-time properties and statistical distributions may differ across different range ambiguous regions. The presence of these

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differences will increase the degrees of freedom and power levels of the clutter, significantly affecting the subsequent performance of multi-channel clutter suppression and moving target detection.

Therefore, several approaches have been proposed to suppress the range ambiguities. Mittermayer [3] proposed a method that involves alternately transmitting up and down chirp signals, allowing the ambiguous energy to propagate in the range direction. This approach significantly reduces the ambiguity peaks and achieves good identification results for point targets. However, since the ambiguous energy is not fundamentally reduced, this method is not suitable for distributed targets. Additionally, multi-angle receiving techniques and digital beamforming (DBF) can be utilized to form high antenna gain equivalent narrow beams directed at the imaging area, which helps to suppress the ambiguous clutter components [4]. However, obtaining narrow beams requires the installation of multiple channels along the elevation dimension, which may lead to substantial data storage and transmission requirements, consuming more system hardware resources.

Considering that in standard pulse-Doppler radar, the pulse signal is transmitted at a fixed pulse repetition interval (PRI), this uniform design exacerbates the phenomenon of range ambiguity. To address this issue, one approach is to adopt an alternating PRF method [5-8], utilizing the Chinese Remainder Theorem, Coincidence algorithm, or Clustering algorithm to design several groups of different pulse interval signals for target detection. Another approach involves altering the same group of pulse intervals. The lengths of different pulse intervals can be pre-defined values [9, 10] or generated randomly [11-14]. Compared to fixed or structured PRI forms, random PRI can produce greater sequence diversity. By randomly deviating from uniform PRI, it effectively suppresses the Doppler ambiguity main lobe caused by repeated occurrences of the same pulses, thereby enhancing ambiguity suppression performance [15].

Based on this, this paper considers applying random pulse interval technology to the suppression of range-ambiguous clutter in spaceborne multi-channel radar and combines it with non-uniform Fourier transform and dimension-reduction space-time adaptive processing techniques for clutter suppression. The performance and applicability of this algorithm are analyzed through simulations of detection probability curves for targets with different yaw angles and varying radar cross sections (RCS).

II. SPACEBORNE RDDAR CLUTTER MODEL

First of all, we establish a spaceborne multi-channel radar clutter model, as shown in Fig. 1. The Z-axis points in the direction opposite to the Earth center, the Y-axis represents the platform motion direction, and the X-axis forms a right-handed coordinate system with the Z-axis and Y-axes. The radar platform height is denoted as H , and the platform velocity is V_a . The origin of the coordinate system is the center of the antenna panel. The receiving azimuth antenna is evenly divided into N equivalent channels, with the physical center distance between adjacent channels being d . The parameters $\theta_{EL,i}$ and $\theta_{AZ,i}$ denote the elevation angle and azimuth angle of the clutter point P_i relative to the radar, respectively. With the consideration of the Earth rotation, the yaw angle is θ_{crab} and the yaw amplitude is ρ_a [16].

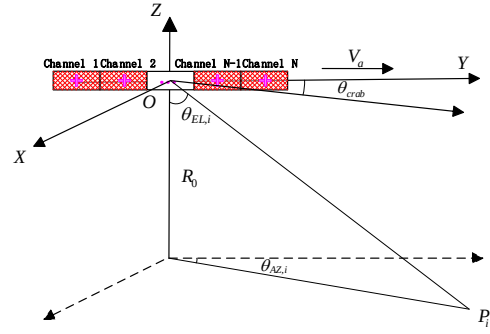


Fig. 2. Three-dimensional coordinates of the clutter point P_i and radar platforms.

In the observation time T_a , assuming the initial slant range $R_0 = \sqrt{x_0^2 + y_0^2 + z_0^2}$ ((x_0, y_0, z_0) is the coordinate of P_i), the instantaneous slant range $R_{n,i}(t)$ between the n th receiving channel and the clutter scattering point P_i at time t_m is expressed as:

$$R_{n,i}(t_m) = \sqrt{\left(\left((2n-1-N)d/2 + V_a t_m \rho_a \cos \theta_{crab} - y_0\right)^2 + \left(V_a t_m \rho_a \sin \theta_{crab} - x_0\right)^2 + z_0^2\right)}. \quad (1)$$

After channel registration of the multi-channel echo data received by the satellite-borne radar, the instantaneous slant range of the scattering point can be simplified as:

$$R_{n,i}(t_m) \approx R_0 - \frac{(2n-1-N)d}{4} \cdot \frac{y_0}{R_0} - V_a t_m \rho_a \left(\frac{y_0}{R_0} \cos \theta_{crab} + \frac{x_0}{R_0} \sin \theta_{crab} \right). \quad (2)$$

The geometric relationship shown in Fig. 1 indicates that $y_0 = R_0 \sin \theta_{EL,i} \cos \theta_{AZ,i}$ and $x_0 = R_0 \sin \theta_{EL,i} \sin \theta_{AZ,i}$. Then (2) can be further simplified as:

$$R_{n,i}(t_m) = R_0 - V_a \rho_a \sin \theta_{EL,i} \cos(\theta_{AZ,i} - \cos \theta_{crab}) t_m - \frac{(2n-1-N)d}{2} \sin \theta_{EL,i} \cos \theta_{AZ,i}. \quad (3)$$

When the next pulse signal is transmitted, the current pulse signal cannot be received, resulting in that pulse signal being received in the subsequent period. This causes the echo signal of the current distance unit to overlap, leading to a range ambiguity phenomenon. Let p represent the index of the range ambiguity, where a positive value indicates a forward range ambiguity region and a negative value indicates a backward range ambiguity region. For the ground scatter point in the p th range ambiguity region, the two-way instantaneous slant range between the p th range ambiguity region and the n th azimuth dimension receiving channel can be expressed as:

$$R_{n,i,p}(t_m) = R_0 - V_a \rho_a \sin \theta_{EL,i} \cos(\theta_{AZ,i} - \cos \theta_{crab})(t_m - p \cdot T_{PRT}) - \frac{(2n-1-N)d}{2} \sin \theta_{EL,i} \cos \theta_{AZ,i} + p \frac{c \cdot T_{PRT}}{2} \quad (4)$$

Assuming that the signal transmitted by the spaceborne radar is a narrowband linear frequency-modulated signal, then the raw echo data received by the n th radar receiver channel after range compression can be given by (5). The gain of the bistatic antenna radiation pattern is represented by G_{TR} , and the backscatter coefficient of the clutter block is represented by $\sigma_{c,i}$. By performing a Fourier transform in the

azimuth dimension on equation (5), the Doppler clutter signal can be expressed as equation (6).

$$s'_{p,n}(t_r, t_m) = \sum_{p=p_1}^{p_2} \sum_{i=1}^{N_c} G_{TR}(t_m) \sigma_{c,i} \text{sinc} \left[B \left(t_r - \frac{2R_{n,i,p}(t_m)}{c} \right) \right] \times \exp \left[-\frac{j4\pi}{\lambda} \left(R_0 - \frac{(2n-1-N)d}{2} \sin \theta_{EL,i} \cos \theta_{AZ,i} \right) \right] \times \exp \left[\frac{j4\pi}{\lambda} V_a \rho_a t_m \sin \theta_{EL,i} \cos(\theta_{AZ,i} - \cos \theta_{crab}) \right] \times \exp \left(-\frac{j2\pi c T_{PRT} p}{\lambda} \right), \quad (5)$$

$$s'_{p,n}(t_r, f_a) = \sum_{p=p_1}^{p_2} \sum_{i=1}^{N_c} \hat{A}_{p,i} \text{sinc} \left[B \left(t_r - \frac{2R_{n,i,p}(f_a)}{c} \right) \right] \times \text{sinc} \left[T_a \left(f_a - \frac{2\rho_a \sin \theta_{EL,i} \cos(\theta_{AZ,i} - \cos \theta_{crab}) V_a}{\lambda} \right) \right] \times \exp \left[-\frac{j4\pi}{\lambda} \left(R_0 - \frac{(2n-1-N)d}{2} \sin \theta_{EL,i} \cos \theta_{AZ,i} \right) \right] \times \exp \left(-\frac{j2\pi c T_{PRT} p}{\lambda} \right), \quad (6)$$

where $\hat{A}_{p,i}$ is the amplitude of the clutter signal in the post-Doppler domain. From Equation (6), it can be seen that when both the yaw angle and range ambiguity are present, the range dependency of the clutter becomes quite severe, which will further deteriorate the performance of multi-channel clutter suppression and target constant false alarm detection. Therefore, to enhance the clutter suppression capability, it is essential to effectively reduce the expansion of clutter spectrum caused by range-ambiguous clutter components.

III. PROPOSED RANGE AMBIGUOUS CLUTTER REJECTION METHOD

In a signal with pulse interval uniformly distributed, the time interval from the n th pulse to the $n+1$ th pulse is represented as t_{PRT} . the method of random pulse repetition interval (PRI) introduces a random component into it, as described by the following formula:

$$t_m = (m-1) \cdot t_{PRT} + \varepsilon_m \cdot t_{PRT}, \quad (7)$$

where ε_m is a random number between $-\alpha/2$ and $\alpha/2$ (α is a number less than $(t_{PRT} - T_p)/t_{PRT}$, and T_p represents the pulse width).

So, the distance difference between the p th range ambiguity and unambiguity corresponding to the m th pulse is:

$$\Delta R_{m,p} = \frac{c}{2} (t_{m+p} - t_m) = \frac{c}{2} (p + \varepsilon_{m+p} - \varepsilon_m) t_{PRT}. \quad (8)$$

According to (8), it can be seen that:

$$\Delta R_{m+1,p} = \frac{c}{2} (p + \varepsilon_{m+1+p} - \varepsilon_{m+1}) t_{PRT} \neq \Delta R_{m,p}. \quad (9)$$

The differences in the p th range ambiguity distance and the unambiguous distance of different pulses are varies. Therefore, the same distance of ambiguity clutter energy being distributed across different pulse distance bins, which does not affect the clutter at the unambiguous region.

Since the slow time t_m is non-uniform, a non-uniform Fourier transform is employed when calculating the post-Doppler frequency, yielding the radar signal Doppler data.

$$S_{p,n}(t_r, f_a) = \sum_{m=0}^{M-1} s'_{p,n}(t_r, t_m) e^{-j \frac{2\pi}{M} t_m f_a}. \quad (10)$$

Based on the radar signal Doppler data, the EFA STAP algorithm is applied to suppress the spaceborne multi-channel clutter [2]. Assuming that three Doppler channels are used for joint processing, for the k th Doppler unit $k = 2, 3, \dots, K-1$, the data vector is composed of data from the $k-1$ th Doppler unit to the $k+1$ th Doppler unit.

$$\tilde{S}_{r,k} = \begin{bmatrix} S_{r,k-1}^T & S_{r,k}^T & \cdots & S_{r,k+1}^T \end{bmatrix}^T, \quad (11)$$

$$S_{r,k} = \begin{bmatrix} S_{r,k,1} & S_{r,k,2} & \cdots & S_{r,k,N} \end{bmatrix}^T, \quad (12)$$

where $S_{r,k,n}$ represents the data from the r th range unit, the k th Doppler unit, and the n th azimuth receiving channel. Thus, the corresponding clutter covariance matrix can be obtained as follows:

$$\tilde{R}_{c,k} = \frac{1}{N_x} \sum_{i=1}^{N_x} \tilde{S}_{i,k} \tilde{S}_{i,k}^H. \quad (13)$$

The clutter suppression weight is:

$$w_k = \frac{\tilde{R}_{c,k}^{-1} \tilde{S}_{i,k} \tilde{S}_{i,k}^H}{\tilde{S}_{i,k}^H \tilde{R}_{c,k}^{-1} \tilde{S}_{i,k} \tilde{S}_{i,k}^H}, \quad (14)$$

where $\tilde{S}_{i,k} = [0 \ 1 \ 0]^T \otimes s_{i,k}$ with $s_{i,k}$ being the target spatial steering vector.

IV. SIMULATION ANALYSIS

The following assesses the impact of using a random PRF on clutter suppression and target detection performance through simulated clutter data from a satellite-based multi-channel early warning radar. The main parameters of the space-based radar system refer to the LLSBR of the United States, as shown in Table 1[17].

TABLE I. RADAR SYSTEM PARAMETERS

platform height	508km
carrier frequency	1.3GHz
signal bandwidth	3MHz
sample frequency	3.6MHz
channel number	16
pulse repetition rate	4000Hz
pulse duration time	40ms
system loss	9dB
noise factor	2dB
downward angle of the beam center	40°
spatial cone angle of the beam center	90°
antenna azimuth dimension	50m

Fig. 3(a) and Fig. 3(b), respectively, show the range-Doppler figures before and after clutter suppression using the random PRI technique. From the figures, it can be observed that the high-speed movement of the satellite radar platform causes significant main-lobe clutter spread. And STAP technology can effectively narrow the clutter spectrum, achieving clutter suppression. Fig. 4(a) and Fig. 4(b) present the results of averaging clutter energy along range bins before and after clutter suppression, comparing no range ambiguity,

classical uniform PRI method and random PRI method with range ambiguity. These figures indicate that while the use of random PRI technology can alleviate the issue of clutter spectrum expansion caused by range ambiguity, it significantly increases the background noise level.

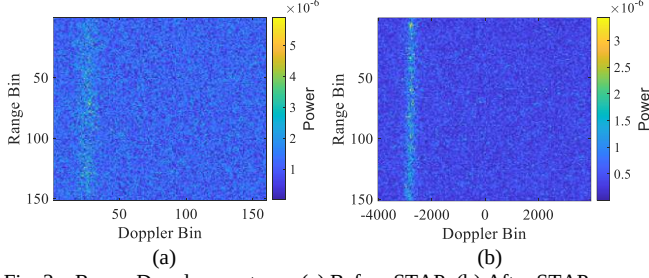


Fig. 3. Range-Doppler spectrum. (a) Before STAP. (b) After STAP.

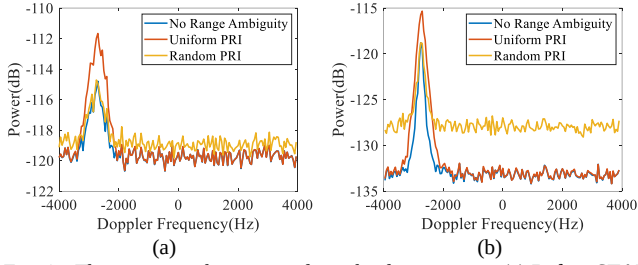


Fig. 4. The average echo energy along the distance unit. (a) Before STAP. (b) After STAP.

Figs. 5(a)(c)(e)(g) show the echo signal-to-clutter and noise ratio (SCNR) curves for different target radial velocities under different yaw angles and different target RCSs. Figs. 5(b)(d)(f)(h) present the target detection probability curves as a function of target radial velocity under different yaw angles and different target RCSs, with a false alarm probability of 1×10^{-6} by using the Swerling 3 model. Table 2 shows the target minimum detectable velocity (MDV) under different yaw angles and target RCS values at detection probabilities of 0.7 and 0.9. It can be observed that, except for the case where the yaw angle is 5 degrees, RCS is 10 m², and the detection probability is 0.9, the MDV index using the random PRI technique deteriorates. In all other conditions, the random PRI technique improves target detection performance, especially when the detection probability is lower and both RCS and yaw angle are larger. This is primarily because, although the use of random PRI technology mitigates the problem of clutter spectrum broadening caused by range ambiguity, the random dispersion of range ambiguous clutter energy increases the radar signal background noise. Therefore, it is particularly suitable in scenarios where the impact of range ambiguity is significant (with larger yaw angles) and the target signal-to-noise ratio (SNR) is relatively high.

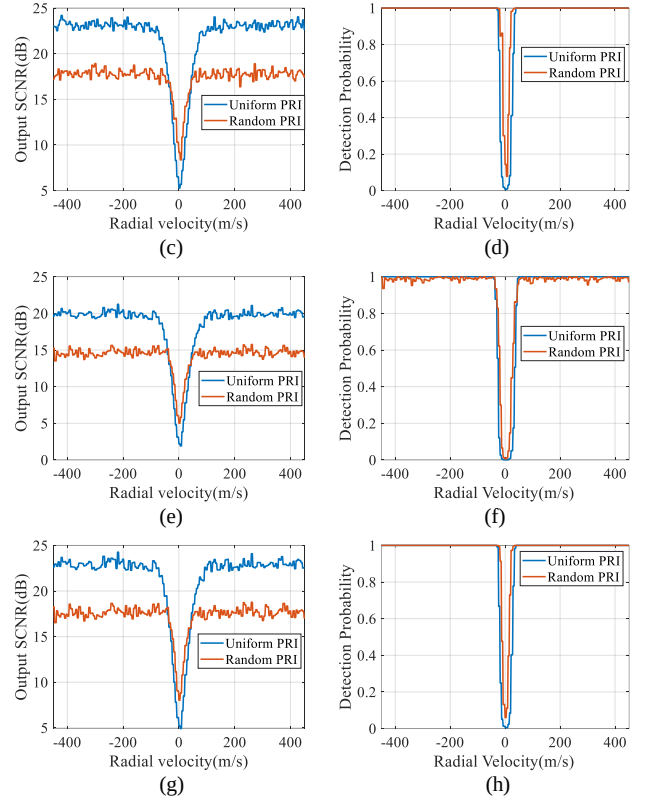
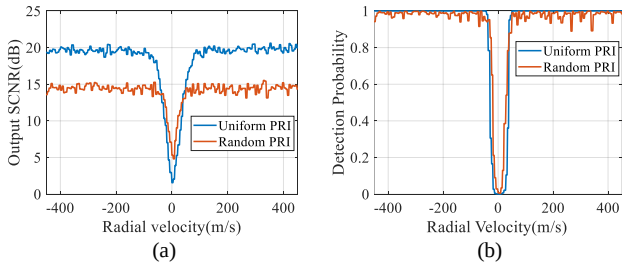


Fig. 5. Target detection performance curve. (a) Target output SCNR diagram of $\theta_{crab}=5^\circ$ and target RCS=10m². (b) Detection probability graph of $\theta_{crab}=5^\circ$ and target RCS=10m². (c) Target output SCNR diagram of $\theta_{crab}=5^\circ$ and target RCS=20m². (d) Detection probability graph of $\theta_{crab}=5^\circ$ and target RCS=20m². (e) Target output SCNR diagram of $\theta_{crab}=10^\circ$ and target RCS=10m². (f) Detection probability graph of $\theta_{crab}=10^\circ$ and target RCS=10m². (g) Target output SCNR diagram of $\theta_{crab}=10^\circ$ and target RCS=20m². (h) Detection probability graph of $\theta_{crab}=10^\circ$ and target RCS=20m².

TABLE II. MINIMUM DETECTABLE VELOCITY OF TARGET UNDER DIFFERENT CONDITIONS

Yaw angle	Target RCS	Detection probability	Uniform PRI MDV	Random PRI MDV	MDV reduction ratio
5°	10m ²	0.7	38.13m/s	35.28m/s	8.08%
		0.9	39.69m/s	40.12m/s	-1.08%
	20m ²	0.7	32.48m/s	18.85m/s	72.31%
		0.9	34.28m/s	20.65m/s	66.01%
10°	10m ²	0.7	38.20m/s	32.28m/s	18.34%
		0.9	43.31 m/s	43.20m/s	0.25%
	20m ²	0.7	29.53m/s	14.44m/s	104.50%
		0.9	32.17 m/s	19.83m/s	62.23%

V. CONCLUSION

This paper studies a range ambiguous clutter suppression method that uses random pulse intervals, combined with non-uniform Fourier transform and EFA STAP techniques to suppress clutter containing range ambiguity. Simulation analysis and validation demonstrate that the random PRI technique is effective but also has limitations, which can alleviate the spectral broadening caused by range-ambiguous signals at the sacrifice of target SNR. Therefore, this method is suitable for moving target detection scenarios with the big yaw angle and the high SNR.

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